Clinical Text Summarization Leveraging Transformer-Based Models: A Comprehensive Survey

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***Abstract*—The increase in volume of unstructured medical records has made it difficult for healthcare practitioners to access important patient data. Extensive clinical notes are often excessively lengthy and inefficient to manually review, especially in critical care settings. We sought to resolve this challenge by creating an AI-powered clinical text summarization model that can generate simple and insightful summaries of discharge reports. The model has been trained on advanced NLP and LLM frameworks, which include OpenAI's O1, DeepSeek R1, and LLaMA, to assess their performance on complex medical text summarisation tasks. This work seeks to evaluate these models in comparison to foundational models of clinical text summarization like BART, T5, PEGASUS, and BART, which were used in the base paper "Clinical Text Summarization using NLP Pretrained Language Models: A Case Study of MIMIC-IV-Notes." We constructed a complete pipeline for summarization by implementing preprocessing steps such as data cleansing, expansion of abbreviations, and pipeline construction via GCP Vertex AI and Ollama. We conduct evaluation through ROUGE benchmark scores to measure summary accuracy. Our hypothesis states that newer models will outperform traditional models in contextual and accurate summary creation. This has the ability to radically transform and reduce load on clinicians.**

I. Introduction

As healthcare record technology advances, medical specialists are increasingly burdened with large amounts of clinical documentation and notes that lack structure. Important details are often mixed with lengthy discharge summaries, which are not easy to interpret. Alongside aiding in automating documentation workflows, tailored summarisation has the potential to simplify mental strains on clinicians, enhance clinical workflows, and improve clinicians' decision quality.

For this project, we focus on solving the clinical text summarisation problem utilising modern NLP frameworks. We use different types of large language models, T5, BART, PEGASUS, LLaMA3, and DeepSeek-R1 model that has been introduced recently. These models focus on summarising documents with clinical notes from the MIMIC-IV-Note dataset alongside clinical corresponding gold reference summaries from the MIMIC-IV-Ext-BHC dataset. To perform rigorous experimentation, we accessed the cloud-based Google Colab and AI Workbench ported with Vertex tools, as well as locally deploying under Ollama for models with higher compute requirements. All models were equally evaluated using standard ROUGE, BLEU, and BERTScore assessment frameworks.

We aim to analyze the traditional and new LLMs in the field of medicine alongside their capabilities of generating accurate, semantically correct, and aligned text summaries. The additional focus of this research involves dealing with medically relevant terms and customizing pretrained model frameworks for the healthcare domain.

II. Literature Review

Arokodare, Wimmer, and Du tested BART, T5, and Pegasus models on the MIMIC-IV-Notes dataset. They found that transformer models perform much better than older approaches for summarizing medical notes. These models work especially well when trained on clinical data. They showed that transformer models can improve both speed and accuracy when summarizing complex medical documents. Their study confirms the importance of using specialized data for training models to summarize clinical text [1]. Chaves, Kesiku, and Garcia-Zapirain reviewed summarization methods in biomedical fields. They found that older machine learning models were commonly used, but deep learning and transformer models performed much better. The newer models improved clarity, medical accuracy, and usefulness in summaries. The study explained that transformer models were especially useful for biomedical texts because of their ability to understand complex

medical language. It shows that deep learning models are now the preferred choice for clinical text summarization [2].

Faray de Paiva and his team tested the DeepSeek R1 model using the USMLE dataset. While DeepSeek R1 performed well on reasoning tasks, it faced some challenges when summarizing clinical text. The model needs more training in medical texts to improve its performance. The study showed that the model could handle general tasks, but it was not yet fully ready for summarizing complex medical documents. Further training on specialized medical data is needed for better results in clinical summarization [3]. Chandra Shekar and his team worked on extractive summarization using convolutional and recurrent neural networks. Their approach performed well in extracting key phrases from clinical documents. However, it did not capture context as well as transformer models. They suggested that the method was useful, but transformer-based models were better at handling complex relationships between different parts of the text. The study shows that transformers are more effective for clinical text summarization where understanding context is important [4].

Lalitha and her team explored abstractive summarization using attention-based neural networks. They used seq2seq with attention mechanisms to improve the quality of summary. Their results showed better summaries than older methods but still not as good as transformer models like BART and T5. The attention-based models were an improvement, but they still haven't achieved the accuracy and coherence of transformer models. This research shows that while attention-based networks are helpful, transformers remain the best choice for clinical summarization [6]. Lu, Ehwerhemuepha, and Rakovski tested deep learning models for classifying medical notes. They found that transformer models worked better than other models, especially when dealing with unstructured data. Medical notes are often messy, and transformers handle this type of data well. Their research showed that transformers could process complex medical information more effectively. This is important for clinical summarization, where data can be incomplete or disorganized sometimes [8].

Schopf, Braun, and Matthes explored unsupervised text classification using zero-shot and similarity-based methods. They focused on how to classify texts without labeled data. While their study was on classification, the findings were relevant for summarization. In clinical settings, there is often a lack of labeled summaries. Their work showed that these unsupervised methods could be useful in testing summarization models when labeled data is rare, which makes it easier to evaluate models in real-world medical settings [9]. Zhang and his team developed a hierarchical transformer for summarization. This model looked at both sentence-level and document-level features to create more organized summaries. By considering the entire document structure, the model produced more coherent summaries. This approach was particularly useful for clinical texts, which are often long and complex. The study showed that hierarchical transformers can improve the organization and clarity of clinical summaries, which is important in medical settings where clarity is essential [11].

Alsentzer and his team introduced ClinicalBERT, a version of BERT trained on clinical data from MIMIC-III. The model performed well on various medical NLP tasks, including summarization. ClinicalBERT was fine-tuned specifically for the medical field, making it more effective than standard BERT for clinical text tasks. The study explained the importance of training models on domain-specific data to improve their performance. It showed that ClinicalBERT was a great choice for summarizing medical texts [5]. Li and his team introduced ChatDoctor, a medical chatbot fine-tuned from the LLaMA model using real patient-doctor conversations. ChatDoctor improved its ability to answer health-related questions and generate clinical responses. By training on actual doctor-patient interactions, the model could provide more accurate and context-aware answers. The research showed the potential of LLaMA-based models for use in clinical settings, especially for providing reliable responses in healthcare environments [7].

Wang and his team proposed LLaMA-Clinic, a smaller version of the LLaMA-2 model trained for generating clinical notes. This model performed well in producing accurate clinical summaries while being more efficient and cost-effective than larger models. LLaMA-Clinic can generate clinical notes at a lower cost, making it ideal for smaller healthcare providers or those with limited computational resources. The study shows that smaller models can still deliver quality results while being more accessible and affordable [10]. All these studies show that transformer models, including those based on LLaMA, are becoming the main choice for summarizing clinical text. There are still some challenges like adapting models to the medical field and making sure they give correct facts. But overall, large language models (LLMs) have great potential to make clinical work easier and save time for doctors.

III. Objectives

The main objective of this project is to test how well transformer-based models can summarize clinical text, such as doctor notes or discharge summaries. We used five models-T5, BART, Pegasus, DeepSeek R1, and LLaMA to see how well each one can create short and accurate summaries from long and complex medical notes. Another objective is to compare the models and understand which ones work best for our clinical text. We also looked at how simple prompt-based methods perform, especially with models like R1 and LLaMA that are good at few-shot learning.

To evaluate the quality of the summaries, we used ROUGE, BLEUscore, and BERTScore. These scores help us measure how similar the model's summaries are to the reference summaries in terms of words, meaning, and structure. Overall, this study aims to show how transformer-based models can help doctors and healthcare workers by saving time and making medical records easier to understand.

IV. Data Collection

The very first step in this project was to find a dataset that is both medical in nature and has a lot of free narrative text, as the dataset was very critical for performing clinical text summarization. Therefore we chose the MIMIC-IV Discharge Notes dataset, which is publicly available on PhysioNet, as it contains some of the most detailed clinical data available. Healthcare workers draft de-identified summaries for patients in critical care units and these summaries are called discharge notes. These notes capture the patient's hospitalization course with all treatments performed, medications given, diagnoses, and discharge instructions. This information is essential needed when building healthcare associated model, thus allowing training and evaluation of summarization model within a healthcare setting.

With the goal of providing dataset diversity while keeping computation requirements reasonable, a subset of 1,000 discharge summaries was selected. MIMIC-IV BHC dataset also contains reference summaries created by human editors. This dataset was supplemented with machine learning classifiers for supervised model training. The dataset contains valuable discharge summaries, discharge summary note identifiers, demographic metadata (anonymized age and gender), diagnosis coded using ICD-9 or ICD-10, medications, as well as timestamps for both admission and discharge.

The different factors contributed to the decision of choosing the MIMIC-IV dataset. First, its clinical context ensured that models developed on this data would encounter real-world medical language in practice, which would improve the summarization output validity. Second, the availability of both structured data such as diagnosis and medication lists as well as narrative text made it possible for the study to use multi-modal data during analysis. Third, the dataset's HIPAA-compliant de-identification procedures made it ethically sound and convenient for academic research, as long as users completed necessary training for data use through PhysioNet's credentialing process. The dataset has several benefits, but the challenges encountered during data preparation include discharge summaries were written by different clinicians which introduced variation in vocabulary, tone, and structure. Furthermore, documents often contained non-standardized medical shorthand, abbreviations, and acronyms which made tokenization and lemmatization difficult.

To resolve these problems, an intensive preprocessing pipeline was put into place. Each discharge summary's text went through a normalization process which erased irregular characters and strange formatting. To lessen the vocabulary, make it more coherent, and especially for clinical terms, lemmatization along with stopword removal was carried out. Cleaned texts were transformed into tokens readable by transformer models through tokenization. Furthermore, when notes surpass the input token threshold, the text is split into smaller chunks without losing context. These formative processes ensured that the data input was clean and structurally adequate for the intended advanced algorithms while also contextually rich for thorough model training and evaluation exercises during the evaluation exercises.

Dataset links : "MIVIC-IV datasets " This link contains the datasets that we have used in this study. You can also access these datasets in "physionet.org" but you will require credential access for accessing these datasets.

V. Exploratory data analysis

The Exploratory Data Analysis (EDA) stage was necessary to understand the underlying structure, content, and issues within the MIMIC-IV-Note dataset, particularly about clinical text summarization. At first glance, it was clear to us that the discharge summaries were quite inconsistent in their lengths, with some being only a few hundred words while others exceeded 5,000 words. The mean word count for the discharge notes is around 1,200, and the reference summaries only average about 120 words. This suggests a compression ratio of approximately 10:1, which is important for model tuning in situations needing abstractive summarization.

For our clinical text summarization project, we extracted the MIMIC-IV discharge summaries, which necessitated considerable cleaning prior to use. This included removing the NaN data which was replaced with "N/A", and changing the placeholder underscores to a more uniform symbol ("**"). Moreover, we normalized excessive white space and tab characters, making everything neater. From there, we further cleaned the text by filtering specific special characters using a regular expression, ensuring that relevant medical punctuation spent unscathed. It was also incredibly important to expand medical abbreviations. Like any other field, clinical shorthand dominated the notes, so we built an abbreviation dictionary and expanded them. This useful change was indispensable for models like PEGASUS, BART, T5, DeepSeek R1, and LLaMA as it allowed them to understand the clinical context better, enhancing the summaries' accuracy and semantics.

A. Hypothesis of study

The expectation is that DeepSeek R1 and LLaMA models will surpass T5, BART, and PEGASUS in clinical text summarisation tasks, which is our main hypothesis for the study. This assumption is based on the claim that the architecture and parameters (DeepSeek R1–32 B, LLaMA3 7B) will better understand complex medical language, contextual understanding, and capture important elements from lengthy discharge summaries. Additionally, we believe that as newer models, LLMs DeepSeek R1 and LLaMA3 will produce summaries more consistent with reference summaries, with higher ROUGE scores as well as superior evaluations for semantic and text coherence, making them more informative and concise, with better alignment to provided reference summaries.

VI. Data Analytics

Data analytics formed the distinctive core focus of this clinical text summarization project, with special attention to the preparatory steps of input data, model choice, training, and performance evaluation. After discharge summaries in the

MIMIC-IV dataset were preprocessed, multiple transformer-based models were trained and tested to create concise and clinically correct summaries. This was accomplished through extensive data preparation followed by iterative refinement using state-of-the-art natural language processing frameworks aimed at abstractive summarization.

The primary goal of the initial analytics phase was to perform the extraction steps such as cleaning and transforming crude clinical notes into a model-ready format that is logical and organized, and free of extraneous details. As discharge notes tend to be verbose and lack a standard format, the dataset also required extensive preprocessing. Special characters, formatting, and punctuation were standardized through removal, and lemmatization reduced words to their base forms. Stop words: simple, non-meaning delivering terms—were omitted to lessen their impact on attention mechanisms. The cleaned text was then split into subword sequences using model-specific tokenizers for each transformer model. The preprocessing pipeline was curated so that the models did not have to deal with anything else apart from medically relevant text, reducing noise and linguistic redundancy.

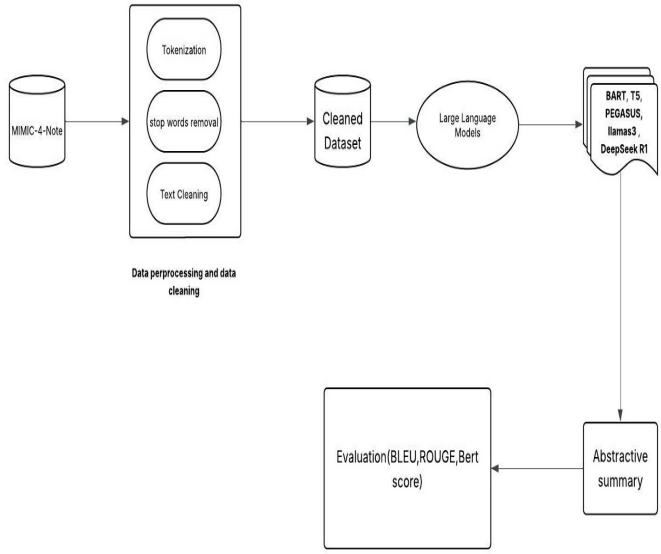

Fig. 1. Research Methodology

After preparing the data, the next step was to choose the appropriate transformers for testing. Initially, three models were selected for their success in general-domain summarization tasks—BART, T5, and PEGASUS. BART was used due to his understanding of bidirectional context and sequence generation using an encoder-decoder structure. T5's fame comes from his text-to-text formulation of all NLP tasks, providing versatility and task relatability. PEGASUS was relevant due to strong long-form text generation performance since he was pre-trained for summarization through a masked sentence prediction strategy. Midway through the project, two more models—LLaMA 3 and DeepSeek R1—were added to test newer instruction-tuned architectures with larger context handling.

Every model was trained on a 1000-record sample taken from the MIMIC-IV dataset. They were subsequently assisted with further training which enabled the models, through domain-specific data, to better fit the clinical context and format of discharge summaries. Each model was optimized on hyperparameters like learning rate, batch size, and epoch count to achieve a middle ground between performance and resource consumption. The model which was held out from the training process was used to assess the generalization ability of the models alongside the validation set and work towards avoiding overfitting the training data.

With respect to evaluating the model performance, integrated performance approach using rigid predefined standards was applied. ROUGE scores were used to evaluate the generated summary versus reference summary overlaps, providing insight into the amount of overlap with the generated summary for reference alongside measuring lexical accuracy. BLEU scores were used to measure the alignment of phrasing with human-drafted summaries which is helpful when text is short and fluent. Semantic similarity evaluation was performed utilizing contextual embeddings in BERTScore measurement which offers a deeper assessment of meaning preservation even in surface-level differencing texts. This range of metrics allowed to evaluate the results on both syntax and semantics.

The analytics process culminated in the stage of results interpretation. The assessment revealed that the latest models, particularly DeepSeek R1 and LLaMA 3, exceeded the baseline models in most evaluation metrics. Given its instruction-tuning and high token capacity, DeepSeek R1 provided strong semantic accuracy and contextual coherence in longer clinical notes. LLaMA 3 garnered praise for its summary readability, effortlessly achieving impressive ROUGE-1 and BLEU scores, marks for fluent expression and structural accuracy, proving it produced well-graded summaries. Meanwhile, BART and PEGASUS delivered satisfactory assessments but lacked full-blown elaboration in their summaries. T5 was found to have the greatest difficulty, particularly with long dense passages of medical terms. These results did not change across validation sets, highlighting the deepseek model's claim of robustness and strong potential in practical clinical summarization tasks.

VII. Results

For this study on clinical text summarization, we have conducted evaluation on discharge notes of 1000 samples of the MIMIC-IV dataset, where the goal was to assess which transformer-based models best suitable for clinical text summarization. Each note was associated with reference summaries that were earlier derived from the MIMIC-IV dataset and which were taken from the MIMIC-IV-BHC dataset. To exactly match the notes from both the datasets, we have mapped them on note id's of both the datasets and took 1000 notes. The models tested included BART, T5, PEGASUS, LLaMA 3, and DeepSeek R1, and they were evaluated using

key performance indicators such as BLEU, ROUGE (1, 2, L), and BERTScore (Precision, Recall, F1).

DeepSeek R1 was the best model across all measures and was able to outperform all the other models. It had the best ROUGE-2 score as well as BERTScore F1, showing mastery in the metrics of semantic coherence and factual retention. Its large context window and instruction tuning allowed for retaining precision while summarizing complex medical narratives with clinical language. LLaMA 3 also achieved remarkable results in ROUGE-1 and BLEU and produced fluent and concise summaries. Relative to other models, its decoder-only architecture enabled it to better preserve contextual relevance while alleviating redundancy.

		BLEU	ROGUE1	ROGUE2	ROGUEL	ROGUELSUM	BERTScore(Precision, Recall, F1)
Model	T5	0.0000006	0.0951	0.0281	0.0693	0.0693	(0.8366, 0.7773, 0.8057)
	Bart	0.0028	0.1187	0.0104	0.064	0.064	(0.7791, 0.7835, 0.7811)
	Pegasus	0.0003	0.1574	0.0676	0.1157	0.1158	(0.8392, 0.7924, 08149)
	llama3	0.0312	0.2947	0.1072	0.1772	0.1828	(0.8577, 0.8168, 0.8365)
	deepseek-r1	0.0235	0.3221	0.074	0.1417	0.2082	(0.8152, 0.8194, 0.8172)

Fig. 2. Research Methodology

Moderate results were yielded by PEGASUS and BART. Although BART was able to formulate readable summaries, some clinically relevant information was neglected. On the other hand, PEGASUS was strong at sentence-level cohesion, but failed to maintain semantic accuracy. T5 struggled with longer inputs, scoring the lowest on most other metrics, indicating undersresourced n dealing with extensive medical documentation. Qualitative analysis of the model outputs confirmed these insights. Summaries generated by DeepSeek R1 and LLaMA 3 were noted to capture salient medical details including diagnosis, procedures, and discharge plans in addition to being framed within the context of the target references, thus affirming the relevance of their surgical precision. These outcomes clearly emphasized the need for trained models specifically aimed at in-depth clinical narratives integrated with EHR data.

VIII. Conclusion

This research analyzed the abstractive summarization performance of five transformer-based language models on one thousand clinical discharge notes from the MIMIC-IV dataset. As expected, results show that both DeepSeek R1 and LLaMA 3 surpassed baseline models BART, PEGASUS, and T5 in both qualitative and quantitative measures. DeepSeek R1's instruction-tuned design and context window length allowed it to produce factually accurate summaries that were rich in substance. LLaMA 3 also performed well regarding fluent and contextually appropriate outputs. Moreover, all these models surpassed others in achieving ROUGE, BLEU, and BERTScore which demonstrates their effectiveness in abstracting salient pieces of clinically pertinent information embedded in verbose medical notes.

The clinical ramifications of this work are tremendous, especially in areas where speedy and reliable access to patient summaries is necessary. These models' effectiveness in producing high-quality summaries could alleviate clinician burden, enhance information flow, and facilitate timely decision-making. Despite the evaluation being confined to a sample of 1,000 records, the results still reveal a significant opportunity to scale these models for use in practical settings. The future work should include broadening the dataset and adding human-in-the-loop assessment as well as incorporating these summarization models into the hospital EHR systems. The quality and generalizability of these models across various clinical venues could be improved by examining ensemble or hybrid approaches.

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